

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool : Debugging & Optimisation Task (Low)

Assessment Tool Weightage: 10%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.	BL2 – Understand
2	Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue. Bloom's Level:	BL2 – Understand
3	Correct a faulty SELECT query where column names are misspelled or improperly referenced.	BL1 – Remember
4	Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.	BL3 – Apply

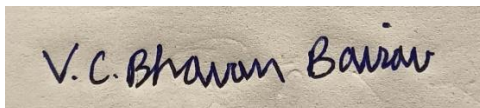
5	Identify and correct a subquery error where multiple values are returned instead of a single value.	BL2 – Understand
6	Debug a VIEW creation statement that contains invalid column references or syntax errors.	BL1 – Remember
7	Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.	BL3 – Apply
8	Identify violations of integrity constraints in a given table and correct them.	BL2 – Understand
9	Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies.	BL3 – Apply
10	Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.	BL3 – Apply

Code of Conduct:

I V.C.Bhavan Bairav - 192511369 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



V.C. Bhavan Bairav

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1) Identify and correct errors in a given SQL CREATE TABLE statement with incorrect syntax and missing constraints.

INCORRECT SQL STATEMENT:

```
mysql> CREATE TABLE Student
-> (
-> StudentID INT,
-> StudentName VARCHAR(50
-> Department VARCHAR(50),
-> Email VARCHAR(100),
-> GPA DECIMAL(3,2)
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for
Email VARCHAR(100),
GPA DECIMAL(3,2)
)' at line 5
mysql>
```

ERRORS:

- 1) Missing) after VARCHAR(50).
- 2) Missing PRIMARY KEY constraint.
- 3) No NOT NULL constraint for mandatory fields.

CORRECT SQL STATEMENT:

```
mysql> CREATE TABLE Student
-> (
-> StudentID INT PRIMARY KEY,
-> StudentName VARCHAR(50) NOT NULL,
-> Department VARCHAR(50) NOT NULL,
-> Email VARCHAR(100) UNIQUE,
-> GPA DECIMAL(3,2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Student VALUES
-> (101,'Aryan','Computer Science','aryan@gmail.com',8.9),
-> (102,'Rahul','Information Technology','rahul@gmail.com',8.2),
-> (103,'Priya','Computer Science','priya@gmail.com',9.1);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Department | Email | GPA |
+-----+-----+-----+-----+-----+
| 101 | Aryan | Computer Science | aryan@gmail.com | 8.90 |
| 102 | Rahul | Information Technology | rahul@gmail.com | 8.20 |
| 103 | Priya | Computer Science | priya@gmail.com | 9.10 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

2) Debug an SQL query that incorrectly uses aggregate functions without GROUP BY and fix the issue.

INCORRECT SQL STATEMENT:

```
mysql> SELECT Department, AVG(GPA)
      -> FROM Student;
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of S
l_group_by
mysql> █
```

ERRORS:

1. AVG(GPA) is an aggregate function.
2. Department is a non-aggregate column.
3. Since Department is selected, it **must** be included in the GROUP BY clause.

CORRECT SQL STATEMENT:

```
mysql> SELECT Department,
      ->      AVG(GPA) AS Average_GPA
      -> FROM Student
      -> GROUP BY Department;
+-----+-----+
| Department          | Average_GPA |
+-----+-----+
| Computer Science    | 9.000000    |
| Information Technology | 8.200000    |
+-----+-----+
2 rows in set (0.00 sec)
```

3) Correct a faulty SELECT query where column names are misspelled or improperly referenced.

INCORRECT SQL STATEMENT:

```
mysql> SELECT S.StudentName, D.DepartmentName
-> FROM Student S
-> JOIN Department D
-> ON S.DeptID = D.DeptID;
ERROR 1054 (42S22): Unknown column 'S.DeptID' in 'on clause'
mysql> SELECT StudntName, Departmnt
-> FROM Student;
ERROR 1054 (42S22): Unknown column 'StudntName' in 'field list'
mysql> _
```

ERRORS:

1. S.DeptID is an incorrect column reference. It should be S.DepartmentID.
2. D.DeptID is an incorrect column reference. It should be D.DepartmentID.
3. StudntName is misspelled. It should be StudentName.
4. Departmnt is misspelled. It should be Department.

CORRECT SQL STATEMENT:

```
mysql> SELECT S.StudentName,
-> D.DepartmentName
-> FROM Student S
-> JOIN Department D
-> ON S.Department = D.DepartmentName;
+-----+-----+
| StudentName | DepartmentName |
+-----+-----+
| Aryan       | Computer Science |
| Rahul       | Information Technology |
| Priya       | Computer Science |
+-----+-----+
3 rows in set (0.01 sec)

mysql> SELECT StudentName, Department
-> FROM Student;
+-----+-----+
| StudentName | Department |
+-----+-----+
| Aryan       | Computer Science |
| Rahul       | Information Technology |
| Priya       | Computer Science |
+-----+-----+
3 rows in set (0.00 sec)
```

- 4) Fix errors in an SQL query using JOIN, where the join condition is missing or incorrect.

INCORRECT SQL STATEMENT:

```
mysql> SELECT S.StudentName,  
->      D.DepartmentName  
-> FROM Student S  
-> JOIN Department D  
-> ON S.DepartmentID = D.DepartmentID;  
ERROR 1054 (42S22): Unknown column 'S.DepartmentID' in 'on clause'  
mysql> █
```

ERRORS:

- 1) **DepartmentID** does not exist in the **Student** table.
- 2) The **Student** table contains the column **Department**, not **DepartmentID**.
- 3) The JOIN condition references a non-existent column, resulting in:
ERROR 1054 (42S22): Unknown column 'S.DepartmentID' in 'on clause'.

CORRECT SQL STATEMENT:

```
mysql> SELECT S.StudentName,  
->      D.DepartmentName  
-> FROM Student S  
-> JOIN Department D  
-> ON S.Department = D.DepartmentName;  
+-----+-----+  
| StudentName | DepartmentName |  
+-----+-----+  
| Aryan      | Computer Science |  
| Rahul      | Information Technology |  
| Priya      | Computer Science |  
+-----+-----+  
3 rows in set (0.00 sec)
```

5) Identify and correct a subquery error where multiple values are returned instead of a single value.

INCORRECT SQL STATEMENT:

```
mysql> SELECT StudentName
-> FROM Student
-> WHERE Department =
-> (
->     SELECT DepartmentName
->     FROM Department
-> );
ERROR 1242 (21000): Subquery returns more than 1 row
mysql>
```

ERRORS:

- 1) The subquery returns **multiple values** (Computer Science, Information Technology).
- 2) The = operator expects **only one value**.
- 3) This results in the error:

ERROR 1242 (21000): Subquery returns more than 1 row

CORRECT SQL STATEMENT:

```
mysql> SELECT StudentName
-> FROM Student
-> WHERE Department IN
-> (
->     SELECT DepartmentName
->     FROM Department
-> );
+-----+
| StudentName |
+-----+
| Aryan       |
| Rahul       |
| Priya       |
+-----+
3 rows in set (0.00 sec)
```


6) Debug a VIEW creation statement that contains invalid column references or syntax errors.

INCORRECT SQL STATEMENT:

MySQL 8.0 Command Line Client

```
mysql> CREATE VIEW Student_View AS
-> SELECT StudentID,
->         Student_Name,
->         Dept
-> FROM Student;
ERROR 1054 (42S22): Unknown column 'Student_Name' in 'field list'
mysql> █
```

ERRORS:

1. Student_Name is an invalid column name. The correct column is StudentName.
2. Dept is an invalid column name. The correct column is Department.
3. Due to invalid column references, the view cannot be created.

CORRECT SQL STATEMENT:

```
mysql> CREATE VIEW Student_View AS
-> SELECT StudentID,
->         StudentName,
->         Department
-> FROM Student;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Student_View;
+-----+-----+-----+
| StudentID | StudentName | Department |
+-----+-----+-----+
| 101 | Aryan | Computer Science |
| 102 | Rahul | Information Technology |
| 103 | Priya | Computer Science |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

7) Examine an SQL query with unnecessary conditions and optimize it by simplifying the WHERE clause.

INCORRECT SQL STATEMENT:

```
mysql> SELECT *
-> FROM Student
-> WHERE Department = 'Computer Science'
-> AND GPA > 8.0
-> AND GPA > 7.0;
```

StudentID	StudentName	Department	Email	GPA
101	Aryan	Computer Science	aryan@gmail.com	8.90
103	Priya	Computer Science	priya@gmail.com	9.10

2 rows in set (0.00 sec)

ERRORS:

1. The condition `GPA > 7.0` is unnecessary because `GPA > 8.0` already satisfies it.
2. Redundant conditions make the query less efficient and reduce readability.

CORRECT SQL STATEMENT:

```
mysql> SELECT *
-> FROM Student
-> WHERE Department = 'Computer Science'
-> AND GPA > 8.0;
```

StudentID	StudentName	Department	Email	GPA
101	Aryan	Computer Science	aryan@gmail.com	8.90
103	Priya	Computer Science	priya@gmail.com	9.10

2 rows in set (0.00 sec)

```
mysql>
```

8) Identify violations of integrity constraints in a given table and correct them.

INCORRECT SQL STATEMENT:

```
mysql> CREATE TABLE Student (  
-> StudentID INT PRIMARY KEY,  
-> StudentName VARCHAR(50) NOT NULL,  
-> Department VARCHAR(50),  
-> Email VARCHAR(100) UNIQUE,  
-> GPA DECIMAL(3,2)  
-> );  
Query OK, 0 rows affected (0.07 sec)  
  
mysql> INSERT INTO Student VALUES  
-> (101,'Aryan','Computer Science','aryan@gmail.com',8.9);  
Query OK, 1 row affected (0.01 sec)  
  
mysql>  
mysql> INSERT INTO Student VALUES  
-> (101,'Rahul','Information Technology','rahul@gmail.com',8.2);  
ERROR 1062 (23000): Duplicate entry '101' for key 'student.PRIMARY'  
mysql>  
mysql> INSERT INTO Student VALUES  
-> (102,NULL,'Computer Science','priya@gmail.com',9.1);  
ERROR 1048 (23000): Column 'StudentName' cannot be null  
mysql>  
mysql> INSERT INTO Student VALUES  
-> (103,'Priya','Computer Science','aryan@gmail.com',9.0);  
ERROR 1062 (23000): Duplicate entry 'aryan@gmail.com' for key 'student.Email'  
mysql>
```

ERRORS:

- **Primary Key Violation**

StudentID = 101 is duplicated.

- **NOT NULL Constraint Violation**

StudentName cannot be NULL.

- **UNIQUE Constraint Violation**

Email aryan@gmail.com already exists.

CORRECT SQL STATEMENT:

```
MySQL 8.0 Command Line Client
mysql> INSERT INTO Student VALUES
-> (101,'Aryan','Computer Science','aryan@gmail.com',8.9);
ERROR 1062 (23000): Duplicate entry '101' for key 'student.PRIMARY'
mysql>
mysql> INSERT INTO Student VALUES
-> (102,'Rahul','Information Technology','rahul@gmail.com',8.2);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO Student VALUES
-> (103,'Priya','Computer Science','priya@gmail.com',9.1);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO Student VALUES
-> (104,'Neha','Computer Science','neha@gmail.com',9.0);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Student;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Department | Email | GPA |
+-----+-----+-----+-----+-----+
| 101 | Aryan | Computer Science | aryan@gmail.com | 8.90 |
| 102 | Rahul | Information Technology | rahul@gmail.com | 8.20 |
| 103 | Priya | Computer Science | priya@gmail.com | 9.10 |
| 104 | Neha | Computer Science | neha@gmail.com | 9.00 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

9) Fix a relation that violates 1NF or 2NF by removing repeating groups or partial dependencies.

INCORRECT SQL STATEMENT:

```
mysql> CREATE TABLE Student_Course (  
-> StudentID INT,  
-> StudentName VARCHAR(50),  
-> Course VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> INSERT INTO Student_Course VALUES  
-> (101,'Aryan','DBMS, Java'),  
-> (102,'Rahul','Python'),  
-> (103,'Priya','DBMS, Java, Python');  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> SELECT * FROM Student;  
ERROR 1146 (42S02): Table 'animal.student' doesn't exist  
mysql> SELECT * FROM Student_Course;  
+-----+-----+-----+  
| StudentID | StudentName | Course |  
+-----+-----+-----+  
| 101 | Aryan | DBMS, Java |  
| 102 | Rahul | Python |  
| 103 | Priya | DBMS, Java, Python |  
+-----+-----+-----+  
3 rows in set (0.00 sec)
```

ERRORS:

1. The Courses column stores **multiple values**.
2. The relation violates **First Normal Form (1NF)** because each attribute should contain only **one value**.
3. Searching, updating, and deleting individual courses becomes difficult.

CORRECT SQL STATEMENT:

```
mysql> CREATE TABLE Student_Course_1NF (  
->     StudentID INT,  
->     StudentName VARCHAR(50),  
->     Course VARCHAR(30)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> INSERT INTO Student_Course_1NF VALUES  
-> (101,'Aryan','DBMS'),  
-> (101,'Aryan','Java'),  
-> (102,'Rahul','Python'),  
-> (103,'Priya','DBMS'),  
-> (103,'Priya','Java'),  
-> (103,'Priya','Python');  
Query OK, 6 rows affected (0.01 sec)  
Records: 6  Duplicates: 0  Warnings: 0  
  
mysql> SELECT * FROM Student_Course_1NF;  
+-----+-----+-----+  
| StudentID | StudentName | Course |  
+-----+-----+-----+  
|      101 | Aryan      | DBMS   |  
|      101 | Aryan      | Java   |  
|      102 | Rahul      | Python |  
|      103 | Priya      | DBMS   |  
|      103 | Priya      | Java   |  
|      103 | Priya      | Python |  
+-----+-----+-----+  
6 rows in set (0.00 sec)
```

10) Identify redundant attributes in a relation and apply basic normalization (up to 3NF) to improve design.

INCORRECT SQL STATEMENT:

```
mysql> CREATE TABLE Student_Course (  
-> StudentID INT,  
-> StudentName VARCHAR(50),  
-> CourseID VARCHAR(10),  
-> CourseName VARCHAR(50),  
-> FacultyName VARCHAR(50),  
-> FacultyPhone BIGINT  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> INSERT INTO Student_Course VALUES  
-> (101,'Aryan','C101','DBMS','Dr. Kumar',9876543210),  
-> (101,'Aryan','C102','Java','Dr. Meena',9876543220),  
-> (102,'Rahul','C101','DBMS','Dr. Kumar',9876543210);  
Query OK, 3 rows affected (0.01 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> select * from student_course;  
+-----+-----+-----+-----+-----+-----+  
| StudentID | StudentName | CourseID | CourseName | FacultyName | FacultyPhone |  
+-----+-----+-----+-----+-----+-----+  
| 101 | Aryan | C101 | DBMS | Dr. Kumar | 9876543210 |  
| 101 | Aryan | C102 | Java | Dr. Meena | 9876543220 |  
| 102 | Rahul | C101 | DBMS | Dr. Kumar | 9876543210 |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```

ERRORS:

1. FacultyName is repeated for the same course.
2. FacultyPhone is repeated multiple times.
3. The relation contains **redundant attributes**, causing:
 - Data redundancy
 - Update anomaly
 - Insert anomaly
 - Delete anomaly
4. The relation is **not in Third Normal Form (3NF)**.

CORRECT SQL STATEMENT:

```
mysql> SELECT * FROM Student;
```

StudentID	StudentName
101	Aryan
102	Rahul

```
2 rows in set (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM Faculty;
```

FacultyID	FacultyName	FacultyPhone
F101	Dr. Kumar	9876543210
F102	Dr. Meena	9876543220

```
2 rows in set (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM Course;
```

CourseID	CourseName	FacultyID
C101	DBMS	F101
C102	Java	F102

```
2 rows in set (0.00 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM Enrollment;
```

StudentID	CourseID
101	C101
102	C101
101	C102

```
3 rows in set (0.00 sec)
```